

DISCLAIMER

This is a **hypothetical** fault isolation procedure created by Nathan Smith solely for demonstration purposes. It contains **no classified, export-controlled, or proprietary information**. All references to schematics, engineering data, test results, part numbers, and procedures are fictional and synthesized for illustrative use only. This document is not for use on actual equipment.

INTERACTIVE ELECTRONIC TECHNICAL MANUAL (IETM) S1000D COMPLIANT – SIMPLIFIED TECHNICAL ENGLISH (STE) UNITED STATES AIR FORCE STYLE

FAULT ISOLATION PROCEDURE (FIP)

TITLE: HIGH VIBRATION ALARM – ISOLATION AND CORRECTION

APPLICABILITY: Solar Turbines Centaur-class Gas Turbine Generator Set (hypothetical model CTS-40-GT)

TASK NUMBER: 71-00-00-810-801 (FIP)

ISSUE DATE: Hypothetical (for demonstration)

REVISION: 1.0

1. PURPOSE

This procedure isolates the cause of a High Vibration alarm using a logical decision tree. It synthesizes data from:

- Engine Control System Schematic (fictional drawing 123-456-789, Rev B)
- Performance Data Sheet PDS-001 (normal vibration limits)
- Sensor Calibration Test Report TR-045 (baseline and acceptance criteria)

2. SAFETY WARNINGS

WARNING: SHUT DOWN THE ENGINE AND ISOLATE FUEL AND ELECTRICAL SOURCES BEFORE ANY MAINTENANCE. FAILURE TO COMPLY CAN CAUSE DEATH OR INJURY.

CAUTION: USE ONLY APPROVED ANTI-VIBRATION MOUNTS AND BALANCING EQUIPMENT. IMPROPER HANDLING CAN CAUSE FURTHER DAMAGE TO THE ROTATING ASSEMBLY.

NOTE: All steps require two-person verification. Record all test results in the maintenance data system.

3. TOOLS AND EQUIPMENT REQUIRED

- Digital vibration analyzer (NSN 4920-01-123-4567, calibrated per TR-045)
- Multimeter (NSN 6625-01-234-5678)

- Laptop with diagnostic interface software (version 5.2 or higher)
- Balance weights kit (P/N 789-012-345)
- Personal Protective Equipment (hearing and eye protection)

4. FAULT ISOLATION DECISION TREE

Follow the steps in sequence. At each decision point, perform the test and branch as directed.

STEP 1 – VERIFY THE ALARM (Reference: Schematic 123-456-789, Sheet 4, Vibration Monitoring Circuit)

1.1 Connect the diagnostic laptop to the engine control unit (ECU).

1.2 Read the fault code and live vibration data.

Recorded peak velocity must be compared to PDS-001 limits (maximum allowable 0.5 inches per second (ips) at running speed).

1.3 Does the alarm clear when the engine is shut down and restarted?

- **YES** → Go to STEP 2 (Sensor or wiring fault probable).
- **NO** → Go to STEP 3 (Mechanical fault probable).

STEP 2 – ISOLATE SENSOR / WIRING FAULT (Reference: Test Report TR-045, Sensor Calibration Data)

2.1 Perform continuity and resistance check on each vibration sensor (Sensors A, B, and C per schematic).

Acceptable resistance: 500 ± 50 ohms (from TR-045).

2.2 If any sensor is open, shorted, or out of tolerance:

- Replace the defective sensor (P/N 456-789-012).
- Re-calibrate per TR-045.
- Return to STEP 1 and verify.

2.3 If all sensors test good:

- Inspect wiring harness for chafing or loose connectors (schematic reference).
- Repair or replace harness as required.
- Go to STEP 4.

STEP 3 – CHECK MECHANICAL SOURCES (Reference: PDS-001, Rotor Balance and Bearing Data)

3.1 With engine shut down and locked out, perform visual inspection of:

- Inlet and exhaust ducts (foreign object damage).
- Coupling alignment (must be within 0.002 in TIR per PDS-001).

3.2 Mount the vibration analyzer and perform a slow-roll check (engine barred over by hand).

- If vibration exceeds 0.1 ips at slow roll:
 - Proceed to bearing inspection and possible replacement (P/N 234-567-890).
- If slow-roll vibration is acceptable:

- Go to STEP 4 (dynamic imbalance or aerodynamic fault).

STEP 4 – RUN ENGINE AT IDLE AND PERFORM SPECTRUM ANALYSIS (Reference: TR-045, Baseline Test Results)

4.1 Start engine and stabilize at idle (per PDS-001).

4.2 Capture vibration spectrum (1×, 2×, and blade-pass frequencies).

- If 1× running speed dominates (>70 % of total amplitude):
 - **YES** → Rotor imbalance confirmed → Schedule field balancing per balance kit instructions (P/N 789-012-345).
 - **NO** → If blade-pass frequency is high → Inspect compressor blades for damage or fouling.
 - 4.3 If vibration remains high after balancing or blade cleaning:
- Replace the entire gas producer rotor assembly (P/N 901-234-567).
- Perform post-maintenance test run and record new baseline data per TR-045.

STEP 5 – FINAL VERIFICATION

5.1 Run the engine through full load cycle.

5.2 Confirm vibration levels remain below 0.5 ips (PDS-001).

5.3 Clear fault code in ECU.

5.4 Update maintenance records with all test data, part numbers replaced, and references used.

END OF PROCEDURE

If the fault cannot be isolated using this decision tree, contact the next higher maintenance level and provide all recorded data from Steps 1–5.

REFERENCES USED IN THIS PROCEDURE:

- Schematic 123-456-789 (Engine Control and Monitoring)
- Performance Data Sheet PDS-001 (Operating Limits)
- Test Report TR-045 (Sensor Calibration and Baseline Vibration)